

Step 3: Theory-Cleaned Supervised PLS and Variable Family Tournament

Step 3 uses the Step 1 VIP evidence and Step 2 redundancy diagnostics to build a more defensible supervised dimension-reduction model. The goal is not to pretend the predictors are independent. The goal is to keep a smaller set of representatives so no single social concept gets double-weighted in the PLS components.

Full vs Cleaned Model

Model	Predictors	Components	CV R2	CV RMSE	Train R2
Full unfiltered PLS	70	12	0.565	0.057	0.693
Theory-cleaned PLS	27	5	0.558	0.058	0.625

Selection Logic

The cleaned model was built with a family approach. Within each family, variables were judged using four pieces of evidence: common-sense interpretability, bivariate turnout association, full-model PLS VIP, and redundancy/VIF. Where variables were close substitutes, the workflow also ran one-family substitution tests and connected-family combination tests.

This means a variable could be excluded even if it had a high VIP when another selected variable represented the same concept more cleanly. Conversely, a variable could stay with moderate VIP if it gives a clearer theoretical reading and does not create severe redundancy.

The 5 selected components are latent variables created from the 27 selected original predictors. They are not 5 named raw variables. The model is interpreted by looking back at which original predictors have high VIP scores and strong loadings within those latent components.

Final Variables Kept

Variable	Family	VIP	Turnout corr	VIF	Reason
block1_age_65_plus_share	age_structure	1.219	0.182	146.305	Kept as a theory/com mon-sense representative and checked against diagnostics.
block1_age_18_34_share	age_structure	0.932	-0.136	129.804	Kept as a theory/com mon-sense representative and checked against diagnostics.
block4_federal_margin	federal_competitiveness	1.065	0.272	8.264	Kept as a theory/com mon-sense representative and checked against diagnostics.
block1_bachelors_or_higher_25_64_share	household_class	1.985	0.554	7.119	Kept as a theory/com mon-sense representative and checked against diagnostics.
block1_average_household_size	household_class	1.834	-0.485	26.655	Kept as a theory/com mon-sense representative and checked against diagnostics.
block1_low_income_lim_at_share	household_class	0.919	-0.199	10.691	Kept as a theory/com mon-sense representative and checked against diagnostics.

Variable	Family	VIP	Turnout corr	VIF	Reason
block2_apartment_share	housing_form_density	0.863	0.089	30.687	Kept as a theory/com mon-sense representative and checked against diagnostics.
block2_condo_share	housing_form_density	0.664	0.084	28.586	Kept as a theory/com mon-sense representative and checked against diagnostics.
block2_population_density_per_km2	housing_form_density	0.744	0.102	47.932	Kept as a theory/com mon-sense representative and checked against diagnostics.
block2_renter_share	housing_tenure	0.671	-0.081	8100.055	Kept as a theory/com mon-sense representative and checked against diagnostics.
block3_visible_minority_share	immigration_citizenship	2.286	-0.642	16.311	Kept as a theory/com mon-sense representative and checked against diagnostics.
block3_non_citizen_share	immigration_citizenship	1.534	-0.417	23.912	Kept as a theory/com mon-sense representative and checked against diagnostics.
block3_recent_immigrant_share	immigration_citizenship	1.227	-0.317	13.072	Kept as a theory/com mon-sense representative and checked against diagnostics.
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.907	-0.535	39.781	Kept as a theory/com mon-sense representative and checked against diagnostics.
block4_mayoral_top_two_margin	mayoral_competitiveness	1.151	0.288	inf	Kept as a theory/com mon-sense representative and checked against diagnostics.
block4_provincial_margin	provincial_competitiveness	0.802	0.184	4.606	Kept as a theory/com mon-sense representative and checked against diagnostics.
block2_same_address_5yr_share	residential_stability	0.644	-0.120	14.819	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_shelter_access_1200m	service_access	0.728	0.139	2.882	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_library_access_1200m	service_access	0.630	0.137	4.646	Kept as a theory/com mon-sense representative and checked against diagnostics.

Variable	Family	VIP	Turnout corr	VIF	Reason
block5_community_centre_access_1200m	service_access	0.495	-0.037	2.138	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.933	0.157	3.371	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_social_housing_share	service_contact	0.747	-0.191	2.642	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_requests_311_per_1000	service_contact	0.765	0.140	10.169	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_development_applications_2021_2025_per_1000	service_contact	0.690	0.095	5.890	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_school_age_5_17_share	service_contact	0.882	-0.194	46.384	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_no_car_household_share	transportation_accesses	0.832	0.174	inf	Kept as a theory/com mon-sense representative and checked against diagnostics.
block5_transit_commute_share_preferred	transportation_accesses	0.538	-0.002	inf	Kept as a theory/com mon-sense representative and checked against diagnostics.

In plain language, the kept variables are age 65 plus share, age 18 to 34 share, bachelor or higher education share, average household size, low income share, renter share, same address five year share, apartment share, condo share, population density, visible minority share, non citizen share, recent immigrant share, effective mayoral candidates above five percent, mayoral top two margin, federal margin, provincial margin, no car household share, transit commute share, shelter access within 1200 metres, library access within 1200 metres, community centre access within 1200 metres, KSI collisions per 1000 residents, social housing share, 311 requests per 1000 residents, development applications per 1000 residents, school age children share.

Cleaned Model VIP Results

Variable	Family	VIP	Turnout corr	Direction
block3_visible_minority_share	immigration_citizenship	2.145	-0.642	lower turnout
block1_bachelors_or_higher_25_64_share	household_class	1.847	0.554	higher turnout
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	1.787	-0.535	lower turnout
block1_average_household_size	household_class	1.672	-0.485	lower turnout
block3_non_citizen_share	immigration_citizenship	1.433	-0.417	lower turnout
block3_recent_immigrant_share	immigration_citizenship	1.137	-0.317	lower turnout
block4_mayoral_top_two_margin	mayoral_competitiveness	1.084	0.288	lower turnout
block4_federal_margin	federal_competitiveness	0.967	0.272	higher turnout

Variable	Family	VIP	Turnout corr	Direction
block1_age_18_34_share	age_structure	0.833	-0.136	lower turnout
block5_school_age_5_17_share	service_contact	0.820	-0.194	higher turnout
block1_age_65_plus_share	age_structure	0.805	0.182	higher turnout
block1_low_income_lim_at_share	household_class	0.770	-0.199	lower turnout
block5_no_car_household_share	transportation_access	0.741	0.174	lower turnout
block5_shelter_access_1200m	service_access	0.716	0.139	lower turnout
block4_provincial_margin	provincial_competitiveness	0.675	0.184	higher turnout
block5_social_housing_share	service_contact	0.674	-0.191	lower turnout
block2_apartment_share	housing_form_density	0.669	0.089	higher turnout
block2_population_density_per_km2	housing_form_density	0.609	0.102	higher turnout
block5_ksi_collision_events_2021_2025_per_1000	service_contact	0.592	0.157	higher turnout
block2_same_address_5yr_share	residential_stability	0.575	-0.120	lower turnout

Remaining VIF After Cleaning

Variable	Family	Cleaned VIF
block1_average_household_size	household_class	17.521
block2_apartment_share	housing_form_density	14.575
block2_renter_share	housing_tenure	14.327
block1_age_18_34_share	age_structure	12.753
block3_non_citizen_share	immigration_citizenship	12.265
block5_school_age_5_17_share	service_contact	10.666
block2_condo_share	housing_form_density	10.022
block3_recent_immigrant_share	immigration_citizenship	9.402
block2_same_address_5yr_share	residential_stability	9.301
block3_visible_minority_share	immigration_citizenship	7.847
block1_low_income_lim_at_share	household_class	6.224
block1_age_65_plus_share	age_structure	5.412
block5_no_car_household_share	transportation_access	5.102
block4_effective_mayoral_candidates_5pct	mayoral_competitiveness	4.677
block1_bachelors_or_higher_25_64_share	household_class	3.890

The cleaned model removes the exact duplicate/inverse variables from the full universe, so the impossible VIFs mostly disappear. Some VIF values remain above 10 because the social geography itself is bundled: household size, age, tenure, apartment form, citizenship, and school-age composition still move together across Toronto CTs. That remaining collinearity should be handled by reading the PLS components and VIP scores, not by treating each coefficient as a standalone causal estimate.

The age variables are a good example of remaining conceptual correlation. Age 18 to 34 share and age 65 plus share are partly compositional, but they are not exact inverses. I kept both because they represent different turnout stories: younger-adult concentration and older-adult concentration. Still, the VIF table shows that age structure remains bundled with household size, residential stability, and school-age composition, so these should be interpreted as a family rather than isolated causal coefficients.

Reference Categories For The Cleaned Model

The cleaned model suggests five main reference categories for the final story: immigration and racialized geography, education and class resources, age and household composition, housing tenure and urban form, and electoral competitiveness. The most stable variables behind these categories are visible minority share, bachelor or higher education share, effective mayoral candidates above

five percent, average household size, non citizen share, recent immigrant share, mayoral top two margin, federal margin, age 18 to 34 share, school age children share, age 65 plus share, low income share. These categories are a way to bridge the model-first workflow back to the kind of reference categories Zack was emphasizing.

Family Tournament Highlights

Family/test	Best candidate	CV R2	CV RMSE
age_structure	block1_age_18_34_share	0.559	0.058
age_structure+immigration_citizenship	block1_age_65_plus_share; block3_visible_minority_share	0.564	0.057
federal_competitiveness	block4_federal_margin	0.558	0.058
household_class	block1_bachelors_or_higher_25_64_share	0.558	0.058
housing_form_density	block2_apartment_share	0.559	0.058
housing_tenure	block2_renter_share	0.558	0.058
housing_tenure+housing_form_density	block2_renter_share; block2_apartment_share	0.559	0.058
immigration_citizenship	block3_visible_minority_share	0.566	0.057
mayoral_competitiveness	block4_effective_mayoral_candidates_5pct	0.560	0.058
mayoral_competitiveness+federal_competitiveness+provincial_competitiveness	block4_effective_mayoral_candidates_5pct; block4_federal_margin; block4_provincial_margin	0.560	0.058
provincial_competitiveness	block4_effective_provincial_parties_5pct	0.562	0.058
residential_stability	block2_same_address_1yr_share	0.558	0.058
service_access	block5_shelter_count_1200m	0.557	0.058
service_contact	block5_ksi_collision_events_2021_2025_per_1000	0.569	0.057
transportation_access	block5_transit_commute_share_preferred	0.561	0.058
transportation_access+service_access+service_contact	block5_no_car_household_share; block5_shelter_count_1200m; block5_ksi_collision_events_2021_2025_per_1000	0.578	0.056

Variable-by-Variable Decisions

- block1_age_65_plus_share (age_structure): Kept. VIP 1.219, turnout correlation 0.182, VIF 146.305. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_age_18_34_share (age_structure): Kept. VIP 0.932, turnout correlation -0.136, VIF 129.804. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_median_age (age_structure): Not kept. VIP 1.268, turnout correlation 0.232, VIF 28.099. Excluded to keep the family from double-weighting a correlated concept.
- block1_age_35_64_share (age_structure): Not kept. VIP 0.708, turnout correlation 0.142, VIF 44.191. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_federal_margin (federal_competitiveness): Kept. VIP 1.065, turnout correlation 0.272, VIF 8.264. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_federal_party_count_5pct (federal_competitiveness): Not kept. VIP 0.912, turnout correlation 0.206, VIF 5.005. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_federal_vote_fragmentation (federal_competitiveness): Not kept. VIP 0.436, turnout correlation -0.057, VIF 163.980. Excluded because it is near-duplicate/inverse/compositional with selected family information.

- block4_effective_federal_parties_5pct (federal_competitiveness): Not kept. VIP 0.416, turnout correlation -0.029, VIF 154.937. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block1_bachelors_or_higher_25_64_share (household_class): Kept. VIP 1.985, turnout correlation 0.554, VIF 7.119. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_average_household_size (household_class): Kept. VIP 1.834, turnout correlation -0.485, VIF 26.655. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_low_income_lim_at_share (household_class): Kept. VIP 0.919, turnout correlation -0.199, VIF 10.691. Kept as a theory/common-sense representative and checked against diagnostics.
- block1_unemployment_rate_share (household_class): Not kept. VIP 1.934, turnout correlation -0.532, VIF 3.556. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_apartment_share (housing_form_density): Kept. VIP 0.863, turnout correlation 0.089, VIF 30.687. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_condo_share (housing_form_density): Kept. VIP 0.664, turnout correlation 0.084, VIF 28.586. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_population_density_per_km2 (housing_form_density): Kept. VIP 0.744, turnout correlation 0.102, VIF 47.932. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_semi_detached_share (housing_form_density): Not kept. VIP 0.570, turnout correlation 0.095, VIF 6.727. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_condos_per_km2 (housing_form_density): Not kept. VIP 0.579, turnout correlation 0.071, VIF 13.166. Excluded to keep the family from double-weighting a correlated concept.
- block2_detached_share (housing_form_density): Not kept. VIP 0.742, turnout correlation -0.047, VIF 25.246. Excluded to keep the family from double-weighting a correlated concept.
- block2_condominium_dwellings_count (housing_form_density): Not kept. VIP 0.571, turnout correlation 0.071, VIF 31.740. Excluded to keep the family from double-weighting a correlated concept.
- block2_apartments_per_km2 (housing_form_density): Not kept. VIP 0.673, turnout correlation 0.115, VIF 69.113. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_apartment_lt5_storeys_count (housing_form_density): Not kept. VIP 1.034, turnout correlation 0.275, VIF inf. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_apartment_duplex_count (housing_form_density): Not kept. VIP 0.866, turnout correlation -0.169, VIF inf. Excluded to keep the cleaned model interpretable and lower dimensional.
- block2_apartment_total_count (housing_form_density): Not kept. VIP 0.634, turnout correlation 0.096, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_structural_type_total_dwellings (housing_form_density): Not kept. VIP 0.543, turnout correlation 0.083, VIF 284.495. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_condo_status_total_dwellings (housing_form_density): Not kept. VIP 0.542, turnout correlation 0.083, VIF 259.010. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_apartment_5plus_storeys_count (housing_form_density): Not kept. VIP 0.566, turnout correlation 0.035, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_renter_share (housing_tenure): Kept. VIP 0.671, turnout correlation -0.081, VIF 8100.055. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_owner_share (housing_tenure): Not kept. VIP 0.666, turnout correlation 0.079, VIF 8033.196. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block3_visible_minority_share (immigration_citizenship): Kept. VIP 2.286, turnout correlation -0.642, VIF 16.311. Kept as a theory/common-sense representative and checked against diagnostics.
- block3_non_citizen_share (immigration_citizenship): Kept. VIP 1.534, turnout correlation -0.417, VIF 23.912. Kept as a theory/common-sense representative and checked against diagnostics.
- block3_recent_immigrant_share (immigration_citizenship): Kept. VIP 1.227, turnout correlation -0.317, VIF 13.072. Kept as a theory/common-sense representative and checked against diagnostics.
- block3_immigrant_share (immigration_citizenship): Not kept. VIP 2.031, turnout correlation -0.573, VIF 34.659. Excluded to keep the family from double-weighting a correlated concept.

- block3_non_official_mother_tongue_share (immigration_citizenship): Not kept. VIP 1.751, turnout correlation -0.491, VIF 21.937. Excluded to keep the family from double-weighting a correlated concept.
- block3_citizen_adult_share (immigration_citizenship): Not kept. VIP 1.401, turnout correlation 0.353, VIF 9.215. Excluded to keep the family from double-weighting a correlated concept.
- block3_english_french_knowledge_share (immigration_citizenship): Not kept. VIP 0.969, turnout correlation 0.265, VIF 5.821. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_effective_mayoral_candidates_5pct (mayoral_competitiveness): Kept. VIP 1.907, turnout correlation -0.535, VIF 39.781. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_mayoral_top_two_margin (mayoral_competitiveness): Kept. VIP 1.151, turnout correlation 0.288, VIF inf. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_mayoral_vote_fragmentation (mayoral_competitiveness): Not kept. VIP 1.775, turnout correlation -0.495, VIF 55.498. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block4_mayoral_candidate_count_5pct (mayoral_competitiveness): Not kept. VIP 0.584, turnout correlation -0.107, VIF 1.952. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_mayoral_winner_margin (mayoral_competitiveness): Not kept. VIP 1.151, turnout correlation 0.288, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block4_provincial_margin (provincial_competitiveness): Kept. VIP 0.802, turnout correlation 0.184, VIF 4.606. Kept as a theory/common-sense representative and checked against diagnostics.
- block4_provincial_party_count_5pct (provincial_competitiveness): Not kept. VIP 0.688, turnout correlation -0.150, VIF 2.354. Excluded to keep the cleaned model interpretable and lower dimensional.
- block4_effective_provincial_parties_5pct (provincial_competitiveness): Not kept. VIP 0.529, turnout correlation -0.052, VIF 89.222. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block4_provincial_vote_fragmentation (provincial_competitiveness): Not kept. VIP 0.530, turnout correlation -0.061, VIF 99.928. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block2_same_address_5yr_share (residential_stability): Kept. VIP 0.644, turnout correlation -0.120, VIF 14.819. Kept as a theory/common-sense representative and checked against diagnostics.
- block2_same_address_1yr_share (residential_stability): Not kept. VIP 0.748, turnout correlation -0.156, VIF 13.103. Excluded to keep the family from double-weighting a correlated concept.
- block5_shelter_access_1200m (service_access): Kept. VIP 0.728, turnout correlation 0.139, VIF 2.882. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_library_access_1200m (service_access): Kept. VIP 0.630, turnout correlation 0.137, VIF 4.646. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_community_centre_access_1200m (service_access): Kept. VIP 0.495, turnout correlation -0.037, VIF 2.138. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_library_count_1200m (service_access): Not kept. VIP 0.990, turnout correlation 0.246, VIF 4.853. Excluded to keep the family from double-weighting a correlated concept.
- block5_shelter_count_1200m (service_access): Not kept. VIP 0.818, turnout correlation 0.082, VIF 3.889. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_shelter_nearest_m (service_access): Not kept. VIP 0.718, turnout correlation -0.146, VIF 3.357. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_library_nearest_m (service_access): Not kept. VIP 0.680, turnout correlation -0.137, VIF 3.964. Excluded to keep the family from double-weighting a correlated concept.
- block5_park_count_1200m (service_access): Not kept. VIP 0.581, turnout correlation 0.108, VIF 1.800. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_community_centre_nearest_m (service_access): Not kept. VIP 0.558, turnout correlation 0.063, VIF 2.461. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_community_centre_count_1200m (service_access): Not kept. VIP 0.424, turnout correlation -0.040, VIF 1.644. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_park_nearest_m (service_access): Not kept. VIP 0.366, turnout correlation 0.066, VIF 1.264. Excluded to keep the cleaned model interpretable and lower dimensional.

- block5_ksi_collision_events_2021_2025_per_1000 (service_contact): Kept. VIP 0.933, turnout correlation 0.157, VIF 3.371. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_social_housing_share (service_contact): Kept. VIP 0.747, turnout correlation -0.191, VIF 2.642. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_requests_311_per_1000 (service_contact): Kept. VIP 0.765, turnout correlation 0.140, VIF 10.169. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_development_applications_2021_2025_per_1000 (service_contact): Kept. VIP 0.690, turnout correlation 0.095, VIF 5.890. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_school_age_5_17_share (service_contact): Kept. VIP 0.882, turnout correlation -0.194, VIF 46.384. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_requests_311_2023_2025_estimated_count (service_contact): Not kept. VIP 0.844, turnout correlation 0.168, VIF 11.940. Excluded to keep the family from double-weighting a correlated concept.
- block5_development_applications_2021_2025 (service_contact): Not kept. VIP 0.666, turnout correlation 0.120, VIF 3.590. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_ksi_collision_events_2021_2025 (service_contact): Not kept. VIP 0.618, turnout correlation -0.050, VIF 3.028. Excluded to keep the cleaned model interpretable and lower dimensional.
- block5_no_car_household_share (transportation_access): Kept. VIP 0.832, turnout correlation 0.174, VIF inf. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_transit_commute_share_preferred (transportation_access): Kept. VIP 0.538, turnout correlation -0.002, VIF inf. Kept as a theory/common-sense representative and checked against diagnostics.
- block5_tts_overlap_area_m2 (transportation_access): Not kept. VIP 0.649, turnout correlation -0.102, VIF 6.638. Excluded to keep the family from double-weighting a correlated concept.
- block5_tts_no_car_household_share (transportation_access): Not kept. VIP 0.832, turnout correlation 0.174, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.
- block5_tts_transit_trip_share (transportation_access): Not kept. VIP 0.538, turnout correlation -0.002, VIF inf. Excluded because it is near-duplicate/inverse/compositional with selected family information.

Final Interpretation

The cleaned PLS model should be treated as the main supervised dimension-reduction candidate. It keeps turnout supervision through PLS, while reducing the most obvious double-counting problems that appear in the full model. If an inverse pair performs almost identically, the final choice should be explained as interpretive rather than purely predictive. For example, renter share and owner share can encode nearly the same tenure gradient with opposite signs; keeping renter share makes the urban turnout story easier to discuss without implying that owner share is empirically irrelevant.

The remaining ambiguous cases are valuable rather than embarrassing. They show where the data cannot clearly distinguish one representative from another, especially inside housing form, service access, and election-competitiveness families. Those should be reported as sensitivity areas, not hidden.